

A Comparative Study Of 0.1% Ropivacaine With Fentanyl Versus 0.125% Bupivacaine With Fentanyl as Continuous Epidural Infusion In Labour Analgesia

Jayaprakash Kuppuswamy*¹, Arthanareeswaran Shanmugam², Sreeraaghanidhi Karunanithi³

1- Assistant Professor, Department of Anesthesiology, Swamy Vivekanandha Medical College Hospital and Research Institute, Namakkal, Tamil Nadu, India

2- Associate Professor, Department of General Medicine, Swamy Vivekanandha Medical College Hospital and Research Institute, Namakkal, Tamil Nadu, India

3- Assistant Professor, Department of Ophthalmology, Swamy Vivekanandha Medical College Hospital and Research Institute, Namakkal, Tamil Nadu, India
(Affiliated to the Dr. M.G.R. Medical University)

Correspondence author* : Jayaprakash Kuppuswamy

ABSTRACT:

Childbirth pain is a complex, individualized experience, and its management significantly influences the birthing process. Modern labor analgesia, guided by evidence-based practices, emphasizes patient autonomy and comfort. Regional anesthesia, particularly epidural blockade, is considered the gold standard for labor pain control, with ropivacaine showing potential advantages over bupivacaine. A prospective randomized control trial conducted at a hospital in Coimbatore compared the use of ropivacaine and bupivacaine in epidural analgesia for 70 parturients. The study ensured patient demographics were comparable between the groups. Both drugs provided substantial pain relief, with ropivacaine demonstrating less motor block compared to bupivacaine. The study found no significant differences in the duration of labor stages, mode of delivery, or neonatal outcomes between the two groups. These findings contribute to the understanding of labor analgesia, suggesting that ropivacaine, despite being less potent than bupivacaine, can be equally effective. The study supports the ongoing shift towards patient-centered approaches in managing labor pain, emphasizing safety, efficacy, and patient satisfaction.

KEYWORDS: Labor analgesia, Ropivacaine, Bupivacaine, Epidural blockade, Patient-centered childbirth.

INTRODUCTION:

Childbirth pain is a multifaceted and individualized experience that can be both joyful and challenging for women. The management of this pain profoundly influences the birthing experience¹. It is important to acknowledge that not all women necessitate or desire analgesic intervention during labor, emphasizing the need for education on pain relief options. The decision to use pain relief should be patient-driven, informed by proper guidance. The American Society of Anesthesiologists (ASA) and the American College of Obstetricians and Gynecologists (ACOG) assert that labor-induced severe pain should not go untreated when safe interventions are available². Maternal request alone is considered a valid reason for pain relief, barring medical contraindications. Historical perspectives on labor pain relief have been shaped by religious, cultural taboos, myths, and controversies, hindering pain-free labor experiences. However, modern labor analgesia, guided by evidence-based practices, has evolved considerably. The ideal analgesic for labor should be safe, minimally affect labor progression, offer long-lasting relief, and be cost-effective. Regional anesthesia, specifically epidural blockade, is the gold standard for labor pain control, with ropivacaine, a commonly used local anesthetic, showing potential advantages over bupivacaine³. An opioid and local anesthetic injected epidurally produces less motor blockage and a quicker onset of analgesia⁴. Compared to either medication taken alone, the pain relief begins earlier and lasts longer. It permits the administration of both medications at lower concentrations, which lowers the possibility of opioid side effects and systemic toxicity from local anesthetics^{5,6}. During labor, bupivacaine and ropivacaine are frequently administered to give effective epidural analgesia. The dangers of cardiac toxicity and motor blockade—which is linked to maternal discontent and an increase in instrumental deliveries—reduce the usefulness of bupivacaine. More sensory motor differential blockage and a lower risk of systemic toxicity are two benefits of ropivacaine. Comparisons between ropivacaine and bupivacaine for labor analgesia have produced contradictory results^{7,8,9}. This study aims to assess whether ropivacaine offers significant benefits in obstetrical outcomes, pain relief, motor block, and labor characteristics compared to bupivacaine. In summary, the landscape of labor pain management has shifted towards patient

autonomy and comfort, with ongoing research refining techniques to ensure minimal pain and maximal satisfaction during childbirth.

METHODOLOGY:

A prospective randomized control trial, conducted at the Dept. of Obstetrics & Gynaecology, G. Kuppusamy Naidu Memorial Hospital, Coimbatore, involved 70 parturients (35 in each group) from December 2011 to December 2012. Institutional ethics and scientific committee approvals were obtained. Informed consent was obtained after counseling on labor analgesia. Normal singleton pregnancies, age 18–35, ASA status I and II, and cervical dilatation of 3–5 cm during active labor were the inclusion criteria. Contraindications to epidural block, multiple pregnancies, preterm pregnancy, and previous cesarean sections were among the exclusion criteria. Based on an automated randomized list, study groups were assigned at random. An 18G intravenous cannula was inserted, initiating a Ringer lactate solution infusion. The patient assumed either a left lateral or sitting position, with meticulous adherence to aseptic precautions while preparing the lower thoracic and lumbar region. Utilizing 2% lignocaine, the interlumbar space between L1 and L4 was infiltrated. Subsequently, an 18G needle was carefully introduced into the interlumbar space, and the epidural needle, oriented with the bevel facing upward, was meticulously advanced until penetrating the interspinous ligament. The stylet was delicately removed, and a 10ml Loss of Resistance (LOR) syringe, containing either air or saline, was affixed. The needle was methodically advanced, with the epidural space identified by LOR. Cautious aspiration was conducted to ensure the integrity of the duramater. If cerebrospinal fluid (CSF) was aspirated, the needle was judiciously repositioned; otherwise, the LOR syringe was detached, and the depth of the epidural space was recorded. A 20G epidural catheter was threaded through the needle into the epidural space, positioned with 5cm remaining intraspinally. Post-placement, aspiration was meticulously performed to confirm appropriate catheter localization. Subsequent to cleansing and the application of an occlusive dressing, a bacterial filter was affixed to the catheter hub. A test dose of 2% lignocaine with adrenaline was administered via the catheter. In the absence of adverse reactions after 5 minutes, the patient transitioned to a supine position. The study group received a specified bolus dose followed by infusion, adhering to the stipulated protocol.

Table1: Study drugs protocol

Group	Bolus	Infusion
A	6ml of 0.2% Ropivacaine	6-8ml/hr of 0.1% Ropivacaine with 2µg/ml fentanyl
B	6ml of 0.25%Bupivacaine	6-8ml/hr of 0.125% Bupivacaine with 2µg/ml fentanyl

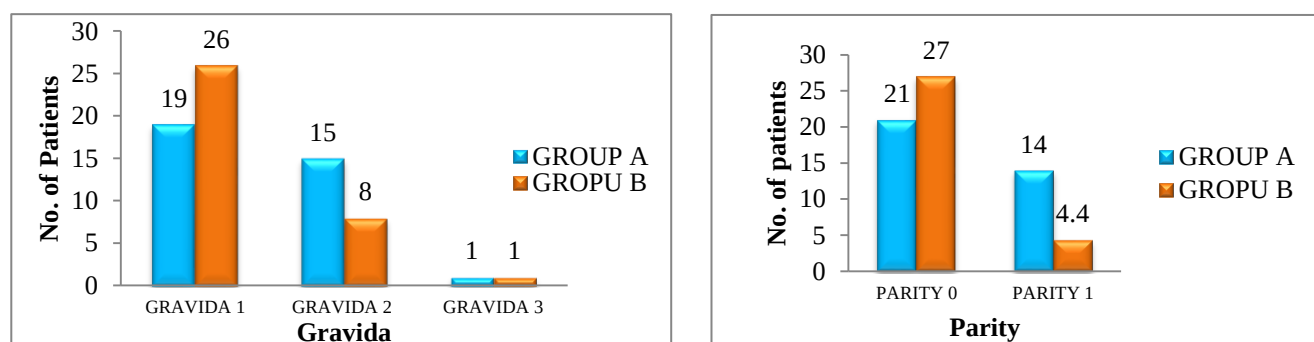
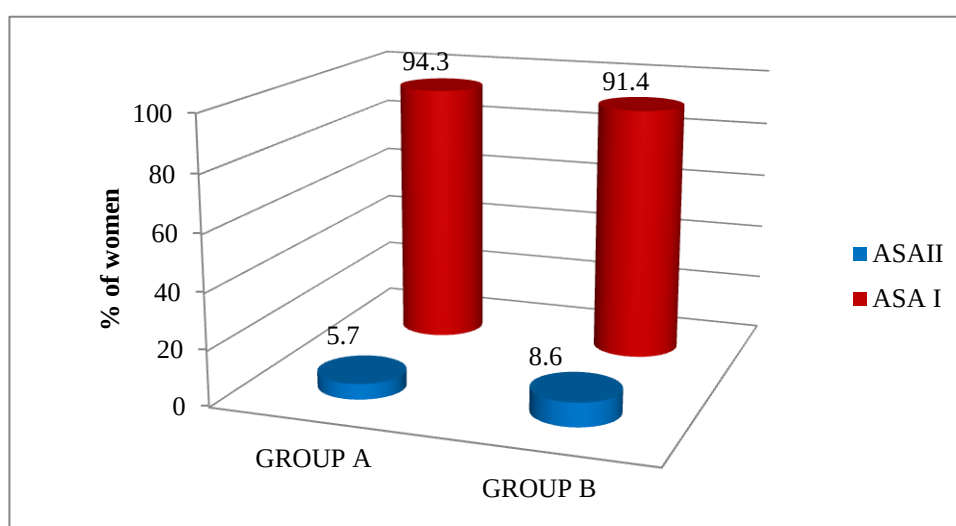
Breakthrough pain was effectively managed by administering 6ml of either 0.2% Ropivacaine or 0.25% Bupivacaine, tailored to the specific study group. To ensure comprehensive observation, maternal parameters were meticulously recorded at 15-minute intervals during the first hour, followed by 30-minute intervals in the second hour, and subsequently on an hourly basis. Concurrently, continuous fetal heart monitoring was diligently conducted, ensuring a thorough assessment of both maternal and fetal well-being. Monitored parameters included maternal heart rate, blood pressure, respiratory rate, and oxygen saturation, alongside pain relief measured by an 11-point Verbal Numerical Rating Scale (VNRS) and motor block assessed by Bromage Score (0-3). Clinical outcomes examined encompassed pain relief, motor block, duration of labor, and mode of delivery (vaginal—spontaneous/assisted, cesarean section). Neonatal outcomes involved APGAR score and NICU admission. Sample size calculation aimed to detect a 40% difference in motor block occurrence, resulting in an optimal sample size of 50 (25 in each group) with 80% power and a 5% level of significance. Statistical analyses, conducted using SPSS version 17 for Windows, involved comparing treatment allocation against case profiles to check for imbalances. Descriptive statistics were presented as mean $\pm$ 1SD, and the Chi-square test for association was employed to compare categorical variables between treatment allocations.

RESULTS:

The age range of patients was 17–36 years, and there was no significant difference in the average age between the two groups. The mean age for women in Group-A was 25.37 ± 3.85 years, while in Group-B. With Gravida and parity of study population the women between two groups were not statistically significant. Out of the 70 pregnant women, 65(92.9%) had ASA grade I pregnancy, the rest 5(7.1%) had ASA grade II pregnancy. The average vaginal dilatation of the whole group was 3.44 ± 0.65 cm. The vaginal dilatation in group-A was 3.37 ± 0.54 cm and in group was 3.37 ± 0.54 .

Table 2 : Demographical parameters of the study population.

Parameters	Group A	Group B
Age (in yrs)	25.37 $\pm$ 3.623	25.23 $\pm$ 3.623
Weight (in kg)	68 8.86	64.29 $\pm$ 9.03

**Fig 1: Gravida and parity of study population****Fig 2: ASA grade of study population.**

In both Group-A and Group-B, one woman (2.9% in each group) had Gestational Diabetes Mellitus (GDM). Pregnancy Induced Hypertension (PIH) was present in one woman (2.9%) in Group-A and two women (5.8%) in Group-B. The distribution of GDM and PIH among the groups was not statistically significant ($p=0.840$). More than 50% of the patients in both the groups A& B received epidural in the L3-4 interspace. Hemodynamics outcomes were measured continuously for 70 patients starting at baseline (before epidural), 15min, 30min, 45min, 1, 1.5, 2, 3, 4, 5, 6, 7 hours. The minimum monitoring time was around 3 hrs in both the groups. The following table will show the number of patients monitored over the time period of their labour.

Table 3: Number of patients at each time point

Time Point	Number of Study population	
	Group-A	Group-B
Baseline	35	35
15 mins	35	35
30 mins	35	35
45 mins	35	35
1 hour	35	35
1.5 hrs	35	35

2 hrs	35	35
3 hrs	35	35
4 hrs	35	33
5 hrs	19	19
6 hrs	10	8
7 hrs	3	3

The following table shows the comparison of heart rate, systolic blood pressure, diastolic blood pressure, respiratory rate between the Groups A&B during their labour. There was no statistically significant difference in the hemodynamics of patients among both groups including heart rate, systolic blood pressure, diastolic blood pressure, respiratory rate among both groups of patients also did not vary significantly.

Table 4: Comparison of mean pain score (VNRS) between group A& B

PARAMETER	TIME	GROUP-A	GROUP-B	t value	p value
PAIN SCORE	Baseline	7.88±0.7	7.65±0.8	1.170	0.246
	15 mins	0.31±0.4	0.17±0.3	1.393	0.168
	30 mins	0.02±0.1	0.08±0.2	-1.023	0.310
	45 mins	0.02±0.1	0.05±0.2	-0.583	0.562
	1 hr	0.02±0.1	0.08±0.2	-1.023	0.310
	1.5 hr	0.11±0.5	0.05±0.2	-1.358	0.179
	2 hr	0.08±0.2	0.02±0.1	1.023	0.310
	3 hr	0.20±0.6	0.08±0.3	1.041	0.302
	4 hr	0.28±0.8	0.09±0.3	1.235	0.221
	5 hr	0.42±0.9	0.52±-.2	-0.289	0.774
	6 hr	0.00±0	0.38±1.1	-1.127	0.276
	7 hr	0.00±0	0.00±0		

There was a noticeable decrease in the pain levels immediately after bolus. The pain levels did not go above VNRS (verbal numerical rating scale) of 3 during infusion in both the groups. Most of the increase in pain scores occurred during the second stage of labour. But the pain score variation did not have any statistical significance. 7 women in both groups required boluses during their labour. The proportion of women requiring boluses was comparable in both the groups. There were more spontaneous vaginal deliveries in Group-A (62.9%) compared to group-B (54.3%). Assisted vaginal deliveries were less in group- A (25.7%) compared to group-B (37.1%). Four patients in group-A (11.4%) and three patients in group-B (8.6%) had cesarean deliveries.

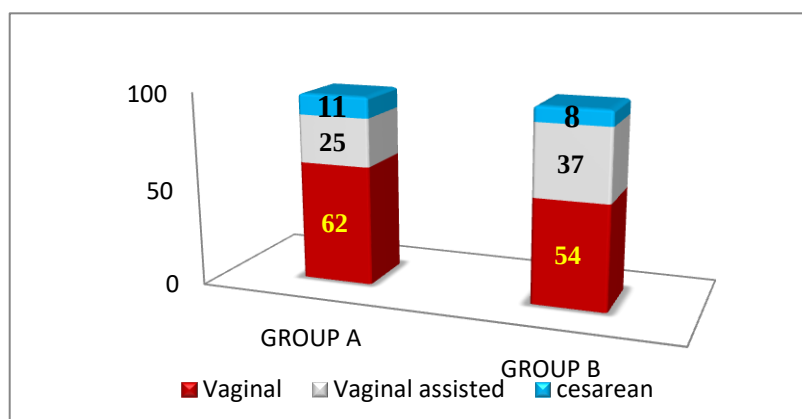


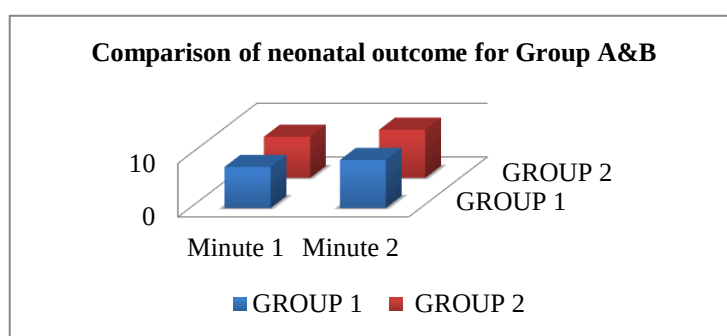
Fig 3: Mode of delivery between group A&B

The following chart shows the average duration of 1st,2nd and 3rd stage of labour in minutes. All 3 stages of labour were comparable.

Table 5: Comparison of Duration of labour between group A&B.

DURATION (IN MINUTES)	GROUP-A	GROUP-B	t value	p value
STAGE-I	467.4±95.8	467.6±87.8	-0.007	0.995
STAGE-II	33.5±8.5	31.1±8.9	1.116	0.269
STAGE-III	6.8±1.7	6.1±1.2	1.769	0.082

The neonatal outcome was rated with APGAR score at 1 & 5 minutes. The average Apgar score during 1st minute assessment was 7.65±0.59 and 7.68±0.47 in group-A and group-B respectively. At 5 minutes, the Apgar score was 8.94±0.23 and 9 in group-A & B respectively.

**Fig 4: Comparison of neonatal outcome by groups**

In the neonatal outcomes, 14.3% of neonates in Group-A and 8.6% in Group-B were admitted to the NICU, with no statistically significant difference ($p=0.845$). NICU admissions There were admissions for cord around the neck, respiratory distress, IUGR, and meconium-stained liquid in Group-B, but there were no admissions for any of these conditions in Group-A. Regarding motor blockade assessed by Bromage score-1, 3 individuals in Group-B experienced this during the 5th hour, with no clinically observable motor blockade in Group-A. However, this difference was not statistically significant ($p=0.071$). Numbness occurred in 2 Group-B patients during the 6th and 7th hours, compared to none in Group-A, but the numbness rate was not statistically significant. Pruritis was not observed in either group.

DISCUSSION :

The Cochrane review⁹, recently published as sheds light on the efficacy of epidural analgesia compared to other labor pain relief methods. According to this review, epidurals provide superior pain relief, reduce the need for additional pain relief, decrease the risk of newborn acidosis, and lower the chances of naloxone administration. Our study aimed to contribute to this knowledge by comparing two widely used local anesthetics, bupivacaine and ropivacaine, for labor epidural analgesia. Bupivacaine, known for its effectiveness in pain relief during labor, was compared with ropivacaine, which offers better sensory-motor differentiation and reduced cardiotoxicity. Several studies have highlighted that ropivacaine's potency is about 60% of bupivacaine^{10,11}. This potency difference was a crucial factor in determining the concentrations used for our labor epidural analgesia. Studies compared equal concentrations of both drugs^{12,13,14,15} or attempted to match their potencies. Most reported comparable pain relief between ropivacaine and bupivacaine, noting less motor block with ropivacaine during extended infusion periods. The recommended doses for labor epidural analgesia involve a range of concentrations and infusion rates. In our study, we initiated labor analgesia with 6 ml of 0.2% ropivacaine (Beillin 1999)¹⁶ based on previous studies and used 6 ml/hour of 0.1% ropivacaine with 2 µg/ml fentanyl (Benhamou 1997, Cascio 1997)^{17,18} for maintenance, in line with prior research. Comparing 0.1% ropivacaine with fentanyl to 0.125% bupivacaine with fentanyl was our primary focus. We ensured comparable demographics between the study groups, including age, weight, gravida, parity, and vaginal dilation. Pain relief, measured through the numerical rating scale (NRS), indicated substantial relief with both ropivacaine and bupivacaine, with no clinically significant differences in onset or overall relief. Similar findings were reported in studies by Meister et al¹⁴. and Fernandez et al¹⁹. Regarding motor blockade, studies have consistently shown less motor block with ropivacaine, a finding supported by our study where ropivacaine had negligible motor block compared to bupivacaine. The study investigated the impact of ropivacaine and bupivacaine on labor duration, mode of delivery, and neonatal outcomes in labor epidural analgesia. The findings showed

no significant differences in the duration of the first stage of labor between the ropivacaine and bupivacaine groups, aligning with previous studies that reported comparable durations. Conversely, studies like Lee et al²⁰ observed a longer first stage of labor with bupivacaine, although they considered the difference clinically limited. Regarding the second stage of labor, the study, consistent with a meta-analysis by Halpern et al²¹, found no disparity in duration between ropivacaine and bupivacaine groups. The study also explored the mode of delivery, highlighting that although the instrumental vaginal delivery rate was lower in the ropivacaine group, the difference was not statistically significant. This finding echoed previous research, indicating similar modes of delivery between both drugs. In terms of cesarean delivery rates, no significant difference emerged between ropivacaine and bupivacaine groups in this study, consistent with previous meta-analyses. Neonatal outcomes, including APGAR scores, showed no significant variation between the two groups, corroborating other studies' conclusions of minimal adverse effects on neonates when using either ropivacaine or bupivacaine for labor analgesia.

CONCLUSION

In conclusion, our study suggests that while ropivacaine is less potent than bupivacaine, it demonstrates comparable efficacy in labor epidural analgesia. This reaffirms that ropivacaine can be as effective as bupivacaine when used in the concentrations employed in our study.

ACKNOWLEDGEMENT

Ramkumar Raghupathy, Dean of G.K.N.M Hospital, for permitting this study in our institution. Special thanks to Dr. Rajani Sundar, Head of the Department of Anaesthesiology, for suggesting the topic and providing constant guidance. I appreciate the valuable guidance of consultants Dr. Soundravalli, Dr. Palaniappan, Dr. Anandhi Arul, Dr. Sai Gopalakrishnan, Dr. Sathyamurthy, and Dr. Muthukumar. Thanks to the Department of Obstetrics and Gynaecology, the labor theatre staff, fellow post-graduates, and Dr. Saleendran for statistical analysis. A heartfelt thanks to my mother, Mrs. Vasanthi, and my wife, Dr. Surya, for their unwavering support. Lastly, gratitude to the patients who participated in this study.

COMPETING INTERESTS We declare no competing interests.

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